

Shivani Ambati

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Education

Marwadi University

Rajkot, Gujarat

B.Tech in Information and Communication Technology — CGPA: 7.85

2023 – 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Digital Signal Processing, Artificial Intelligence, Deep Learning, Object-Oriented Programming

Projects

Computer Vision-Based Robotic Object Sorting System 🤖 | Python, OpenCV, Image Processing

- Developed and deployed a **Python/OpenCV computer vision system for robotic arm sorting**, achieving **88% objects** per minute.
- Implemented **HSV color segmentation, contour detection, and object localization** to accurately identify colored objects in real time.
- Designed the sorting logic to enable a **robotic arm** to classify and pick objects based on detected colors, simulating an automated industrial sorting workflow.

TinyML-Based Seizure Detection and Emergency Alert System 🧠 | Healthcare AI, Edge Impulse, MQTT, Node-RED

- Engineered a **wearable Edge AI platform** by integrating **ESP32-C3**, MPU6050, and MAX30102 sensors for real-time tonic-clonic seizure monitoring, reducing **detection latency by 30%**.
- Deployed a **TinyML model** with Edge Impulse, achieving **94.4% validation accuracy** and a **0.94 weighted F1-score** on seizure classification.
- Architected and implemented a robust IoT communication framework with **MQTT, Node-RED, and Telegram**, ensuring instant monitoring and reducing alert notification latency by **30%**.

RTOS-Based Industrial Vibration Monitoring and Predictive Maintenance System 🧠 | FreeRTOS, TinyML

- Engineered a **real-time vibration analytics platform** on **ESP32** using **FreeRTOS**, separating sensor acquisition, ML inference, alert handling, and dashboard communication into **independent concurrent tasks**.
- Trained and deployed an **Edge Impulse TinyML model** for vibration pattern classification, achieving **89% accuracy** with on-device inference latency below **76 ms** for **predictive fault detection**.
- Developed a **live dashboard using ESP32-C3**, featuring **machine health scoring**, anomaly alerts, and real-time vibration trend visualization for industrial equipment monitoring.

Technical Skills

AI & Machine Learning: TinyML (Edge Impulse), Computer Vision (OpenCV), Image Processing, Data Classification, Predictive Modeling, CNNs

Languages & Frameworks: Python, C/C++, NumPy, Pandas, Scikit-Learn

Software Development: Object-Oriented Programming (OOP), Git, GitHub

Tools: VS Code, Edge Impulse, Node-RED

Achievements

- Solved **100+ LeetCode problems**, ranking among the **top 4.2%** of users.
- Secured a **Top 6** position in the **Byte Hunt IoT Trial** competition.
- Completed **TinyML & Edge AI Training** on embedded AI deployment.
- Completed **VLSI & Nanofabrication Bootcamp** on semiconductor technology and Verilog.

Leadership & Collaboration

Ex-General Secretary, Cloud Computing & DevOps Club 📌

May 2025 – Jan 2026

Led AI, cloud, and DevOps workshops; **40% attendance growth** and **95% satisfaction rating**.

AWS AI & ML Event Coordinator 📌

Aug 2025

Organized AWS UG Rajkot session with **2 industry speakers**.